ModelIt!

Student Activity Pack

What Makes Things Move?

Pushes and Pulls All Around Us

Kindergarten | NGSS K-PS2-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Circle which one is a PUSH: (picture of pushing a door / picture of pulling a door handle)

2 Draw yourself pushing something.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Push Strength | Distance Traveled

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gentle Push

Push Strength set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Big Push

Push Strength set to high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What makes a toy car start moving across the floor? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design a Push-Powered Racer

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YOUR ENGINEERING MISSION

Build a ramp and test how different push strengths make a toy car travel different distances.

THE SCENARIO

The school is holding a Toy Car Race Day! Your team needs to figure out exactly how hard to push your car so it stops right on the finish line — not too far and not too short. Use what you learned about push strength to win!

Engineering Design Thinking

1. WHERE WILL YOU PUT YOUR FINISH LINE?
2. HOW WILL YOU MAKE SURE YOU PUSH THE SAME WAY EACH TIME?
3. HOW WILL YOU MEASURE HOW FAR THE CAR GOES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Car Engineers design how cars move, stop, and steer safely. They use pushes and pulls every day to make cars work! They earn $65,000-$105,000/year.